

CS8803-EMS Project

Email + Residential ISPs

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Motivation

- **How hard is it to send emails from a residential internet connection?**
- Privacy.
 - no one else has the opportunity to look at your messages or advertise alongside them.
 - In an added bonus, your email will keep working even when Gmail, Outlook, or iCloud happens to be down.

Why Hillary Clinton's E-Mail Server Is Less Odd Than You Think

- Spammers and phishers

Approach – Our Approach

- This is largely an experimental study.
- Try to send emails from our residential connection.
 - Go from well configured => poorly configured
 - Record Observations

Approach – How Our Approach is Different

- It probably is not novel, but there hasn't been much recent work in the space where rate of false positive is negligible.
 - Examination based on policy filtering
- It is practical and our results speak volumes for criteria set by the spam filters of prominent email servers.

Results: What Restrictions are applied by the ISP?

- We Tried:
 - Xfinity
 - Google Cloud, AWS*, Azure, Linode
- Port 25 (SMTP + SMTP Exchange) is blocked. A lot.
- Why? Ports 465 and 587 are preferred:
 - Messaging, Malware and Mobile Anti-Abuse Working Group (**M3AAWG**)
 - Internet Engineering Task Force (**IETF**)
- Also restricted (reportedly):



Implementation: Our Test Setup



- Connection: Small Rural Wireless Internet Service Provider (WISP)
 - They don't block Port 25! 😊
 - No IPV6 😞
 - No Port Forwarding 😞
 - No PTR Records 😞
- Server Setup
 - Postfix
 - Dovecot
 - Security Services
 - Webmail



Implementation: DNS + Email

Sender Policy Framework

SPF: Who can send mail for my domain?

```
"v=spf1 ip4:11.22.33.44 -all"
```

Domain-based Message Authentication, Reporting & Conformance

DMARC: What if the email isn't from me?

```
"v=DMARC1; p=reject; sp=none; adkim=s; aspf=s; rua=mailto:postmaster@mail.bvemail.net"
```

DomainKeys Identified Email

DKIM: How do you know the mail is from my server?

```
"v=DKIM1; k=rsa;  
p=MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAuQJZYUagjJ8DC2GoeZBwaE+5ypEx9zjsaDx7xDAJ  
aMVdPnKQ290ibMAcP5VKa05+630E1iKmayKLa7P1V/k
```

Standard Records: A, AAAA, MX

Configured at the Domain (us)

Pointer Record: Reverse DNS

PTR: What domain does this IP belong to?

Configured at the ISP (not us)

Sender Policy Framework

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"v=spf1 ip4:11.22.33.44 -all"
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DKIM: How do you know the mail is from my server?

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"v=DKIM1, k=rsa,  
p=MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAuQJZYUagjJ8DC2GoeZBwaE+5ypEx9zjsaDx7xDAJ  
aMVdPnK9290ibMAcP5VKa05+630E1iKmayKLa7P1V/k
```

Domain-based Message Authentication, Reporting & Conformance

DMARC: What if the email isn't from me?

```
"v=DMARC1; p=reject; sp=none; adkim=s; aspf=s;  
rua=mailto:postmaster@mail.bvemail.net"
```

Standard Records: A, AAAA, MX

Configured at the Domain (us)

Pointer Record: Reverse DNS

PTR: What domain does this IP belong to?

Configured at the ISP (not us)

Implementation: Our Tests



Correct



Not Configured



Incorrect

TESTS	<u>DMARC</u>	<u>DKIM</u>	<u>SPF</u>	<u>Other DNS Records</u>
Test 1				
Test 2				
Test 3				
Test 4				

- Classifying Results
 - Delivered: Inbox or Spam
 - Rejected: Error vs No Error

	<u>DMARC</u>	<u>DKIM</u>	<u>SPF</u>	<u>Other DNS Records</u>
Test 1				
Test 2				
Test 3				
Test 4				



Configured
and Matches



Not Configured



Configured and
Does Not Match

	Test 1	Test 2	Test 3	Test 4
GMAIL	inbox	spam	rejected	rejected
Yahoo	inbox	spam	inbox	spam
Yandex	rejected	rejected	rejected	rejected
Proton	inbox	inbox	inbox	inbox
Outlook	rejected	rejected	rejected	rejected
GMX	rejected	rejected	rejected	rejected
Tutanota	rejected	rejected	rejected	rejected
AOL	spam	inbox	spam	spam

Retrospective

- How was it different/harder than expected?
 - Finding an Internet connection for testing was more difficult than expected.
 - Getting email into inbox was easier than expected.
- What would you do different?
 - Diversify the sending addresses

What remains to be done?

- Conduct a second round of testing with unique email addresses.
- Try AWS unblocking again.

References/Resources

- Xfinity - <https://www.xfinity.com/support/articles/email-port-25-no-longer-supported>
- IETF RFCs
 - <https://www.rfc-editor.org/rfc/rfc5068>
 - <https://www.rfc-editor.org/rfc/rfc4409>
- M3AAWG Paper - https://www.m3aawg.org/sites/default/files/document/MAAWG_Port25rec0511.pdf
- AirFi - <https://www.airfi.today/>
- iRedMail - <https://www.iredmail.org/index.html>
- Google Domains - <https://domains.google.com/registrar/>
- Port Testing - <http://portquiz.net/>
- Email/DNS Checking
 - <https://toolbox.googleapps.com/apps/checkmx/>
 - <https://mxtoolbox.com/emailhealth/mail.bvemail.net/>
 - <https://www.dmarcanalyzer.com/dkim/dkim-checker/>
 - <https://dkimvalidator.com/>
- Gmail Deliverability - <https://support.google.com/mail/answer/81126?hl=en>
- Email Server Configuration - <https://arstechnica.com/information-technology/2014/02/how-to-run-your-own-e-mail-server-with-your-own-domain-part-1/>

Errors - Yandex, Outlook

Undelivered Mail Returned to Sender



From MAILER-DAEMON@8803server.bvemail.net on 2022-11-28 11:42

[Details](#) [Headers](#) [Download all attachments](#)

[Delivery report \(~466 B\)](#) [Undelivered Message \(~2 KB\)](#)

This is the mail system at host 8803server.bvemail.net.

I'm sorry to have to inform you that your message could not be delivered to one or more recipients. It's attached below.

For further assistance, please send mail to postmaster.

If you do so, please include this problem report. You can delete your own text from the attached returned message.

The mail system

<georgiburdell@yandex.com>: host mx.yandex.ru[77.88.21.249] said: 550 5.7.1 Policy rejection on the target address 1669653743-LgSenC2SkGk1-afJjQAtY (in reply to RCPT TO command)

Reporting-MTA: dns; 8803server.bvemail.net
X-Postfix-Queue-ID: 4NLWTm5DMYz2y5K
X-Postfix-Sender: rfc822; gburdell@mail.bvemail.net
Arrival-Date: Mon, 28 Nov 2022 11:42:20 -0500 (EST)

Final-Recipient: rfc822; georgiburdell@yandex.com
Original-Recipient: rfc822;georgiburdell@yandex.com
Action: failed
Status: 5.7.1
Remote-MTA: dns; mx.yandex.ru
Diagnostic-Code: smtp; 550 5.7.1 Policy rejection on the target address 1669653743-LgSenC2SkGk1-afJjQAtY

Subject Test 1: Fully Configured
From gburdell@mail.bvemail.net

Undelivered Mail Returned to Sender



From MAILER-DAEMON@8803server.bvemail.net on 2022-11-28 11:43

[Details](#) [Headers](#) [Download all attachments](#)

[Delivery report \(~747 B\)](#) [Undelivered Message \(~2 KB\)](#)

This is the mail system at host 8803server.bvemail.net.

I'm sorry to have to inform you that your message could not be delivered to one or more recipients. It's attached below.

For further assistance, please send mail to postmaster.

If you do so, please include this problem report. You can delete your own text from the attached returned message.

The mail system

<georgiburdell@outlook.com>: host outlook-com.olc.protection.outlook.com[104.47.18.225] said: 550 5.7.1 Unfortunately, messages from [199.255.149.177] weren't sent. Please contact your Internet service provider since part of their network is on our block list (S3150). You can also refer your provider to <http://mail.live.com/mail/troubleshooting.aspx#errors>. [VI1EUR06FT055.eop-eur06.prod.protection.outlook.com] (in reply to MAIL FROM command)

Reporting-MTA: dns; 8803server.bvemail.net
X-Postfix-Queue-ID: 4NLWVw2cCdZy5M
X-Postfix-Sender: rfc822; gburdell@mail.bvemail.net
Arrival-Date: Mon, 28 Nov 2022 11:43:20 -0500 (EST)

Final-Recipient: rfc822; georgiburdell@outlook.com
Original-Recipient: rfc822;georgiburdell@outlook.com
Action: failed
Status: 5.7.1
Remote-MTA: dns; outlook-com.olc.protection.outlook.com
Diagnostic-Code: smtp; 550 5.7.1 Unfortunately, messages from [199.255.149.177] weren't sent. Please contact your Internet service provider since part of their network is on our block list (S3150). You can also refer your provider to <http://mail.live.com/mail/troubleshooting.aspx#errors>. [VI1EUR06FT055.eop-eur06.prod.protection.outlook.com]

Subject Test 1: Fully Configured

Errors - Google

Undelivered Mail Returned to Sender



From MAILER-DAEMON@8803server.bvemail.net on 2022-11-28 14:55

[Details](#) [Headers](#) [Download all attachments](#)

[Delivery report \(~888 B\)](#) [Undelivered Message \(~2 KB\)](#)

This is the mail system at host 8803server.bvemail.net.

I'm sorry to have to inform you that your message could not be delivered to one or more recipients. It's attached below.

For further assistance, please send mail to postmaster.

If you do so, please include this problem report. You can delete your own text from the attached returned message.

The mail system

```
<burdellgeorge8@gmail.com>: host gmail-smtp-in.l.google.com[64.233.177.26]
said: 550-5.7.25 [199.255.149.177] The IP address sending this message does
not have a 550-5.7.25 PTR record setup, or the corresponding forward DNS
entry does not 550-5.7.25 point to the sending IP. As a policy, Gmail does
not accept messages 550-5.7.25 from IPs with missing PTR records. Please
visit 550-5.7.25 https://support.google.com/mail/answer/81126#ip-practices
for more 550 5.7.25 information.
14-20020a250b0e000000b006f43247ad2csi7162128ybl.410 - gsmt (in reply to
end of DATA command)
```

Reporting-MTA: dns; 8803server.bvemail.net
X-Postfix-Queue-ID: 4NLbmT1bFxz2y5V
X-Postfix-Sender: rfc822; gburdell@mail.bvemail.net
Arrival-Date: Mon, 28 Nov 2022 14:55:21 -0500 (EST)

Final-Recipient: rfc822; burdellgeorge8@gmail.com
Original-Recipient: rfc822;burdellgeorge8@gmail.com

Action: failed

Status: 5.7.25

Remote-MTA: dns; gmail-smtp-in.l.google.com

Diagnostic-Code: smtp; 550-5.7.25 [199.255.149.177] The IP address sending this message does not have a 550-5.7.25 PTR record setup, or the corresponding forward DNS entry does not 550-5.7.25 point to the sending IP. As a policy, Gmail does not accept messages 550-5.7.25 from IPs with missing PTR records. Please visit 550-5.7.25 <https://support.google.com/mail/answer/81126#ip-practices> for more 550 5.7.25 information. 14-20020a250b0e000000b006f43247ad2csi7162128ybl.410 - gsmt

Subject Test 3: No DNS

Undelivered Mail Returned to Sender



From MAILER-DAEMON@8803server.bvemail.net on 2022-11-28 15:24

[Details](#) [Headers](#) [Download all attachments](#)

[Delivery report \(~885 B\)](#) [Undelivered Message \(~2 KB\)](#)

This is the mail system at host 8803server.bvemail.net.

I'm sorry to have to inform you that your message could not be delivered to one or more recipients. It's attached below.

For further assistance, please send mail to postmaster.

If you do so, please include this problem report. You can delete your own text from the attached returned message.

The mail system

```
<burdellgeorge8@gmail.com>: host gmail-smtp-in.l.google.com[64.233.177.26]
said: 550-5.7.26 The MAIL FROM domain [mail.bvemail.net] has an SPF record
with a hard 550-5.7.26 fail policy (-all) but it fails to pass SPF checks
with the ip: 550-5.7.26 [199.255.149.177]. To best protect our users from
spam and phishing, 550-5.7.26 the message has been blocked. Please visit
550-5.7.26 https://support.google.com/mail/answer/81126#authentication
for more 550 5.7.26 information.
g65-20020a252044000000b006e5050ff32fsi10406681ybg.193 - gsmt (in reply to
end of DATA command)
```

Reporting-MTA: dns; 8803server.bvemail.net
X-Postfix-Queue-ID: 4NLcPW6rHHz2y5V
X-Postfix-Sender: rfc822; gburdell@mail.bvemail.net
Arrival-Date: Mon, 28 Nov 2022 15:23:59 -0500 (EST)

Final-Recipient: rfc822; burdellgeorge8@gmail.com
Original-Recipient: rfc822;burdellgeorge8@gmail.com

Action: failed

Status: 5.7.26

Remote-MTA: dns; gmail-smtp-in.l.google.com

Diagnostic-Code: smtp; 550-5.7.26 The MAIL FROM domain [mail.bvemail.net] has an SPF record with a hard 550-5.7.26 fail policy (-all) but it fails to pass SPF checks with the ip: 550-5.7.26 [199.255.149.177]. To best protect our users from spam and phishing, 550-5.7.26 the message has been blocked. Please visit 550-5.7.26 <https://support.google.com/mail/answer/81126#authentication> for more 550 5.7.26 information. g65-20020a252044000000b006e5050ff32fsi10406681ybg.193 - gsmt

Subject Test 4